



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 13.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No. : BITS F234
Course Title : Introduction to Engineering Design
Instructor-in-Charge : ABHIJIT DATE, GEETA BHATT
Practical Instructors : Manjeet Kumar

1. Course Description:

This course provides students with a foundation in engineering design practice and prepares them for discipline-specific advanced courses in engineering design. Students will gain experience in the use of digital design techniques and management of larger open-ended, team-based engineering design tasks in a project-based learning environment. Students will learn to create detailed designs of real-world products and systems by applying design thinking and scientific principles. Students will learn a range of design techniques to contextualize, evaluate, and communicate designs, from generating ideas via sketching to development of digital models and detailed technical drawings that meet the international standards. Students will develop foundation skills in computer-aided design (CAD) techniques to generate 3D models of products and assemblies and learn to integrate them in augmented and virtual reality tools.

2. Course Contents: Engineering Design process, Introduction to Engineering Drawing, Introduction to Orthographic Projections, Projections of points, lines, planes, and solid objects, Sectional views, Manufacturing processes and materials, Fits and Tolerances, Isometric Drawing, Introduction to Solid Modelling, Assembly drawings

3. Scope and Objective of the Course:

The scope and objectives of the course are as follows:

1. Apply key elements of the design process and scientific principles to create detailed design of a product that meets engineering requirements and standards.
2. Generate and interpret engineering technical drawings of parts and assemblies according to engineering design standards; and
3. Use CAD software to generate a computer model of a well-defined part or assembly.

4. Text Books:

Engineering Drawing and Design, 6th Edition David A Madsen, Cengage Learning

5. Reference Books:

1. Engineering Graphics Essentials with AutoCAD 2023, Kirstie Plantenberg, SDC Publications.



6. Course Plan:

Module No.	Topic	Lab Activity	Learning outcomes
1 (Week 1 and 2)	Course overview and Engineering Design process	Group activity - brain storming on innovative design ideas	Principles of sustainable and ethical design.
2 (Week 3)	Introduction to Engineering Drawing	Student will make 2D drawings with free hand and then with AutoCAD.	Tool and methods of presenting/communicating ideas through visual representations.
3 (Week 4 and 5)	Introduction to Orthographic Projections	Student will make 2D drawings with free hand and then with AutoCAD.	Learn to draw the orthographic projections for given objects (pictorial views)
4 (Week 6)	Projections of points, lines, planes, and 3D objects	Student will practice projection of lines and planes with free hand and then with AutoCAD.	Learn to draw projections of lines and planar surfaces placed in different quadrants.
5 (Week 7)	Dimensioning and Tolerancing	Student will practice dimensioning with free hand and then with AutoCAD	Introduction to dimensioning techniques (linear, angular, and radial dimensions), Geometric dimensioning and tolerancing (GD&T) principles
6 (Week 10)	Isometric Projections and 3D CAD	Student will make isometric drawings with free hand and then with AutoCAD. Part modelling: Student will make 3D drawings with AutoCAD and Solid Works. This is to make the student familiarize with AutoCAD 3D interface and Solid Works interface.	Learn to produce isometric, 3D solids modelling drawings
7 (Week 11)	Introduction to assembly and sectional views	Assembly: Students will make 3D drawings (assembly and sectional views) with AutoCAD and Solid Works. This is to familiarize the student with AutoCAD 3D interface and Solid Works interface.	Learn to draw exploded views and bill of materials (BOM). Detailed and exploded assembly views
8 (Week 12)	Manufacturing processes and materials	Students will learn the material selection procedure in commercial software	Learn to link design ideas to manufacturing. Discussion on sustainable manufacturing.
9 (Week 13 onwards)	Project	Students will work on their project during the lab classes	



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7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Closed Book/ Open Book)
Mid-Semester Test	90 Min.	20	Will be announced by IC	Closed Book
Comprehensive Examination	2 h 10 min	30	30/11/2026 FN	Closed Book
Lab Assessments (Online Quizzes)		20	Will be announced by IC	Open Book
Project		30	Evaluation due in week 15	Open Book

8. Chamber Consultation Hour: Room No. 6020-C, Thursday 10:00 AM to 11:00 AM.

9. Notices: Over email/through Canvas.

10. Make-up Policy: Make-up requests must be submitted well in advance.

11. Note (if any):

Instructor-in-charge
BITS F234